

Bending, tension, and thermal membrane fluctuations

A fluctuating membrane provides two different measurements: its equal-time roughness probes elastic costs, while its relaxation probes dissipation. This note derives that distinction for a nearly flat, passive membrane. The aim is a useful baseline for more elaborate membrane models, not a model of all fluctuations in a living cell.

1 Elastic energy and thermal fluctuations

Consider a fluid membrane with bending modulus κ , Gaussian modulus $\bar{\kappa}$, prescribed tension $\sigma \geq 0$, mean curvature H , Gaussian curvature K , and spontaneous curvature C_0 . The Helfrich energy is [1]

$$F = \int dA \left[\frac{\kappa}{2} (2H - C_0)^2 + \bar{\kappa} K + \sigma \right]. \quad (1)$$

The tension here is the coefficient conjugate to area in the chosen harmonic model. Identifying it with a measured mechanical tension requires care beyond that approximation.

Take a symmetric membrane, $C_0 = 0$, with spatially constant moduli. Write its shape as a height $h(\mathbf{x}, t)$ above a periodic square of side L and projected area $A = L^2$. The expansion requires $|\nabla h| \ll 1$. To quadratic order, $dA \simeq [1 + |\nabla h|^2/2] d^2x$ and $2H \simeq \nabla^2 h$, up to the immaterial normal-orientation sign. For this periodic geometry the Gaussian term does not contribute to the height-dependent quadratic energy. Subtracting the flat-state constant gives

$$F = \frac{1}{2} \int_A d^2x [\kappa (\nabla^2 h)^2 + \sigma |\nabla h|^2]. \quad (2)$$

Define $h_{\mathbf{q}} = A^{-1} \int_A h(\mathbf{x}) e^{-i\mathbf{q}\cdot\mathbf{x}} d^2x$, with $\mathbf{q} = 2\pi(n_x, n_y)/L$ and $q = |\mathbf{q}|$. Then

$$h(\mathbf{x}) = \sum_{\mathbf{q}} h_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{x}}, \quad F = \frac{A}{2} \sum_{\mathbf{q} \neq 0} (\kappa q^4 + \sigma q^2) |h_{\mathbf{q}}|^2. \quad (3)$$

Fix the mean height to remove the translation mode $\mathbf{q} = 0$. Because h is real, $h_{-\mathbf{q}} = h_{\mathbf{q}}^*$: opposite wavevectors are not independent complex degrees of freedom.

Equipartition of the independent real sine and cosine amplitudes yields

$$\boxed{\langle |h_{\mathbf{q}}|^2 \rangle = \frac{k_B T}{A(\kappa q^4 + \sigma q^2)}}. \quad (4)$$

Here T is temperature and k_B is Boltzmann's constant. The explicit A is part of the Fourier convention; omitting it changes the inferred modulus. Both terms in the denominator have units of energy per fourth power of length, before multiplication by A .

2 Hydrodynamic relaxation and observable spectra

Varying Eq. (2), integrating by parts, and using periodicity gives the normal restoring force per projected area,

$$f = -\frac{\delta F}{\delta h} = -\kappa \nabla^4 h + \sigma \nabla^2 h, \quad f_{\mathbf{q}} = -(\kappa q^4 + \sigma q^2)h_{\mathbf{q}}. \quad (5)$$

Energy specifies a force, but does not specify a time scale. Assume identical incompressible Newtonian fluids of viscosity η on the two sides, unbounded in the normal direction, negligible inertia, no permeation, and no additional membrane or intermonolayer dissipation. Solving the linear Stokes problem for a normal Fourier mode gives the mobility [2]

$$\partial_t h_{\mathbf{q}} = \Lambda(q) f_{\mathbf{q}}, \quad \Lambda(q) = \frac{1}{4\eta q}. \quad (6)$$

The flow penetrates a distance of order q^{-1} into each fluid; viscous traction therefore scales as $\eta q \partial_t h_{\mathbf{q}}$. The factor four is the result for the stated two-sided geometry. It must not be carried unchanged to a supported membrane or a confined fluid.

Adding equilibrium thermal forcing gives

$$\partial_t h_{\mathbf{q}} = -\Gamma(q) h_{\mathbf{q}} + \xi_{\mathbf{q}}, \quad \Gamma(q) = \frac{\kappa q^3 + \sigma q}{4\eta}, \quad (7)$$

$$\langle \xi_{\mathbf{q}}(t) \xi_{\mathbf{q}'}(t') \rangle = \frac{2k_B T \Lambda(q)}{A} \delta_{\mathbf{q}, -\mathbf{q}'} \delta(t - t'). \quad (8)$$

The Gaussian noise has zero mean. Its covariance is fixed by the requirement that the stationary distribution be the Boltzmann distribution of Eq. (2). Indeed the variance balance is $\partial_t \langle |h_{\mathbf{q}}|^2 \rangle = -2\Gamma \langle |h_{\mathbf{q}}|^2 \rangle + 2k_B T \Lambda/A$, reproducing Eq. (4) at stationarity.

Define the two-sided spectrum by $S_h(q, \omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle h_{\mathbf{q}}(t) h_{-\mathbf{q}}(0) \rangle$. For a stationary mode,

$$\langle h_{\mathbf{q}}(t) h_{-\mathbf{q}}(0) \rangle = \frac{k_B T e^{-\Gamma(q)|t|}}{A(\kappa q^4 + \sigma q^2)}, \quad S_h(q, \omega) = \frac{2k_B T \Lambda(q)/A}{\Gamma(q)^2 + \omega^2}. \quad (9)$$

The integral of S_h over $d\omega/(2\pi)$ returns the equal-time variance. This is a useful normalization check when comparing temporal and spatial measurements.

Crossovers and hidden area. For $\sigma > 0$, bending and tension become comparable at $q = \sqrt{\sigma/\kappa}$. At longer wavelengths the variance scales as q^{-2} and the relaxation rate as q ; at shorter wavelengths they scale as q^{-4} and q^3 . These are model predictions over a range where both the continuum description and the fluid assumptions hold.

The fractional excess area is $\alpha = (\int dA - A)/A \simeq \langle |\nabla h|^2 \rangle / 2$. Replacing the mode sum by a two-dimensional integral gives

$$\alpha \simeq \frac{k_B T}{4\pi} \int_{q_{\min}}^{q_{\max}} \frac{q dq}{\kappa q^2 + \sigma} = \frac{k_B T}{8\pi\kappa} \log \frac{\kappa q_{\max}^2 + \sigma}{\kappa q_{\min}^2 + \sigma}. \quad (10)$$

The lower cutoff reflects the system size, roughly $2\pi/L$. The upper cutoff reflects the spatial resolution or breakdown of the continuum membrane model; it is not an arbitrary fitting

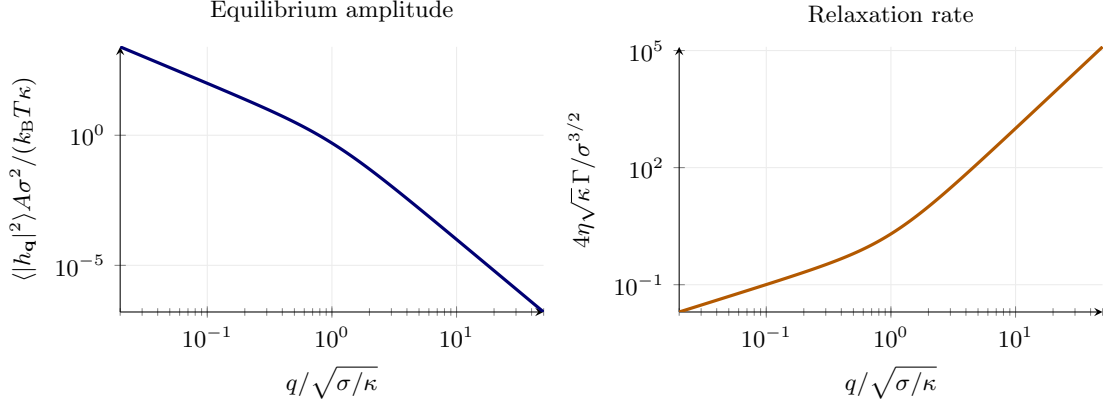


Figure 1: Analytic spectra of the harmonic model, rescaled by the bending–tension crossover. The curves are theoretical, not experimental data. The same crossover has different signatures in equilibrium amplitudes and relaxation rates.

freedom. Increasing tension suppresses fluctuations and releases hidden area without requiring equivalent molecular stretching. The approximation itself requires $2\alpha \ll 1$.

Numerical check. For illustrative values $T = 300$ K, $\kappa = 20k_B T$, $\sigma = 10^{-7}$ N/m, and $\eta = 10^{-3}$ Pa s, the crossover wavelength $2\pi\sqrt{\kappa/\sigma}$ is $5.72 \mu\text{m}$. Equation (7) gives a relaxation time of 0.189 ms at wavelength $1 \mu\text{m}$, and 48.0 ms at $10 \mu\text{m}$. Wavelength means $2\pi/q$, not $1/q$.

What changes in a biological membrane? A height spectrum alone does not identify the dissipation mechanism. Intermonolayer friction introduces an additional slow mode [2]; confinement changes the fluid mobility; and membrane composition can couple to curvature. Neither those effects nor active forcing can be incorporated reliably by changing the bending modulus alone. For a curved, compositionally varying membrane, the coupled balance laws and entropy production provide a systematic route [3].

A practical fitting sequence is to state the Fourier normalization, establish a spatial range with adequate resolution, infer elastic parameters from static data, and then test the dynamics using those parameters. Failure of the single-exponential prediction can be informative. It is not, by itself, proof of activity: passive hidden variables also generate several time scales. In particular, the passive flat-patch model above is not a complete description of a red blood cell and its cytoskeleton.

References

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